ANALYSIS REPORT

Track 1 SVM Final Closure Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the dedicated `SVR` final-closure campaign prepared in:

- `doc/reports/campaign_plans/2026-04-14-20-50-01_track1_svm_final_closure_campaign_plan_report.md`

The campaign executed `12` exact-paper validation runs through the dedicated launcher:

- completed runs: `12`
- failed runs: `0`
- execution window: `2026-04-14 21:09:00+02:00` to `2026-04-14 21:10:39+02:00`
- campaign artifact root:

`output/training_campaigns/track1_svm_final_closure_campaign_2026_04_14_20_50_01/` The campaign goal was explicit:

- close the last residual yellow `SVM` cells after the broader repair wave;
- focus only on amplitude harmonics `40`, `156`, `240`;
- focus only on phase harmonic `162`;
- merge the best per-cell improvements back into the canonical `SVM` row.

Objective And Outcome

The campaign had three concrete goals

1. fully close amplitude harmonic `156`;
2. try to close amplitude `40`, amplitude `240`, and phase `162`;
3. convert the `SVM` row from near-closure into a practically closed row.

Outcome:

- all `12` runs completed successfully;
- all `12` runs exported ONNX artifacts with `0` failed exports;
- the campaign closed amplitude harmonic `156` on both Table `2` and Table `3`;
- the campaign improved phase `162`, but did not fully close it;
- the `SVM` row is now extremely close to full closure, with only `5` yellow

cells left across Tables `2-5`. This means the campaign succeeded partially and usefully:

- it did not finish the full `SVM` row;
- but it removed the strongest remaining amplitude blocker and tightened the

row further without regressions.

Ranking Policy

This campaign is not a full-matrix family comparison. Each run only covers one scoped residual slice of the `SVM` row.

The serialized campaign policy is therefore

- primary metric: `cell_closure_score_desc`
- first tie breaker: `met_paper_cell_count_desc`
- second tie breaker: `open_paper_cell_count_asc`
- third tie breaker: `mean_normalized_gap_ratio_asc`
- fourth tie breaker: `max_normalized_gap_ratio_asc`
- fifth tie breaker: `improved_cell_count_desc`
- sixth tie breaker: `status_upgrade_score_desc`
- seventh tie breaker: `run_name`

Where:

- `cell_closure_score = (met + 0.5 * near) / paper_cell_count`
- `met` means the scoped repository cell reached or beat the paper cell
- `near` means the positive gap stayed within `25%` of the paper cell
- `status_upgrade_score` rewards `red -> yellow`, `yellow -> green`, and `red -> green` transitions versus the pre-final-closure `SVM` baseline row Interpretation:
- this policy exists only to serialize the required campaign-best artifacts;
- the real scientific interpretation remains the merged post-final-closure `SVM` row across Tables `2-5`.

Campaign Ranking

Ranked Completed Runs

Rank	Run	Scope	Cells	Met	Near	Open	Closure Score	Improved	Upgrades
1	track1_svm_amplitude_full_closure_split15	amplitudes_only	6	2	3	1	0.583	3	2
2	track1_svm_amplitude_final_closure_seed23	amplitudes_only	6	1	5	0	0.583	0	0
3	track1_svm_phase_final_closure_split15	phases_only	2	0	2	0	0.500	0	0
4	track1_svm_amplitude_40_bridge	amplitudes_only	2	0	2	0	0.500	0	0
5	track1_svm_phase_final_closure_seed11	phases_only	2	0	2	0	0.500	0	0
6	track1_svm_phase_final_closure_baseline	phases_only	2	0	2	0	0.500	0	0
7	track1_svm_phase_final_closure_split25	phases_only	2	0	2	0	0.500	0	0
8	track1_svm_phase_final_closure_seed23	phases_only	2	0	2	0	0.500	0	0
9	track1_svm_amplitude_final_closure_baseline	amplitudes_only	6	0	5	1	0.417	1	0
10	track1_svm_amplitude_hard_tail_focus	amplitudes_only	4	0	3	1	0.375	1	0
11	track1_svm_amplitude_final_closure_seed11	amplitudes_only	6	0	2	4	0.167	0	0
12	track1_svm_amplitude_hard_tail_seed11	amplitudes_only	4	0	0	4	0.000	0	0

Export Surface

Completed Runs	Exported ONNX Files	Failed Exports	Surrogate Exports
12	22	0	5

All `5` surrogate exports came from amplitude-side `SVR` targets that still needed the existing constant-linear-regression safeguard.

Campaign Best Run

The explicit bookkeeping winner is

- `track1_svm_amplitude_full_closure_split15`

  It was selected because it achieved:
- the highest campaign scoped cell-closure score among runs with a real status

  upgrade;
- `2` met cells and `3` near cells out of `6` ;
- the only successful `yellow -> green` promotion on harmonic `156` across

  both amplitude surfaces;
- the strongest scoped final-closure amplitude improvement package.

  Important interpretation:
- this run is the campaign bookkeeping winner;
- it is **not** the whole scientific result by itself;
- the scientific output is the merged post-final-closure `SVM` row now written

  into the canonical benchmark.

SVM Row Impact

Table-Level Before Vs After

Surface	Before	After
Table `2` amplitude `MAE`	`8` green, `2` yellow, `0` red	`9` green, `1` yellow, `0` red
Table `3` amplitude `RMSE`	`7` green, `3` yellow, `0` red	`8` green, `2` yellow, `0` red
Table `4` phase `MAE`	`8` green, `1` yellow, `0` red	`8` green, `1` yellow, `0` red
Table `5` phase `RMSE`	`8` green, `1` yellow, `0` red	`8` green, `1` yellow, `0` red

Closed Or Improved Cells

- Table `2` amplitude `MAE` : harmonic `156` moved from yellow to green
- Table `3` amplitude `RMSE` : harmonic `156` moved from yellow to green
- Table `4` phase `MAE` : harmonic `162` improved numerically but stayed yellow
- Table `5` phase `RMSE` : harmonic `162` improved numerically but stayed yellow

Residual SVM Yellow Cells

- Table `2` : `40`
- Table `3` : `40` , `240`
- Table `4` : `162`
- Table `5` : `162`

  There are still no remaining red `SVM` cells in Tables `2-5` .

Canonical Benchmark Impact

After merging the best final-closure `SVM` cells into the canonical benchmark:

- Table `2` full-matrix totals moved from `53 / 37 / 10` to `54 / 36 / 10`
- Table `3` full-matrix totals moved from `52 / 35 / 13` to `53 / 34 / 13`
- Table `4` full-matrix totals stayed at `52 / 37 / 1`
- Table `5` full-matrix totals stayed at `43 / 41 / 6`

  The campaign did not change the canonical harmonic-level Table `6` closure reading. Its impact stays local to the `SVM` row and the row-level full-matrix surface.

Main Conclusions

The campaign supports four conclusions.

1. The Hardest Remaining SVM Amplitude Cell Was Closed

Amplitude harmonic `156` is no longer yellow on either paper amplitude surface.

That removes the strongest residual amplitude blocker from the `SVM` row.

2. Phase 162 Improved, But Did Not Close

The best phase run tightened both `MAE` and `RMSE` on harmonic `162`.

However, both cells remain yellow, so the phase row is still not fully closed.

3. SVM Is Now A Micro-Repair Problem

The `SVM` row is no longer a broad repair target.

What remains is a very small residual closure set:

- amplitude `40`
- amplitude `240`
- phase `162`

4. The Best Next Move Is A Tiny Follow-Up Or A Family Shift

At this point, a large new `SVM` campaign would be inefficient.

The rational options are

- one last tiny `SVM` micro-closure pass on `40`, `240`, and `162`; or
- shift the main repair budget to other rows while keeping `SVM` on a minimal follow-up budget.

Produced Campaign Artifacts

The campaign output folder now contains the required serialized winner artifacts:

- `output/training_campaigns/track1_svm_final_closure_campaign_2026_04_14_20_50_01/campaign_leaderboard.yaml`
- `output/training_campaigns/track1_svm_final_closure_campaign_2026_04_14_20_50_01/campaign_best_run.yaml`
- `output/training_campaigns/track1_svm_final_closure_campaign_2026_04_14_20_50_01/campaign_best_run.md`

The canonical reports updated by this closure are:

- `doc/reports/analysis/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`